

Anticancer Drugs

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III-Alkylating agents

- They are CCNS *cellcyclenonspecific*
- *All are mutagenic and carcinogenic, leading to a second malignancy as acute leukemia.*
- A -Mechlorethamine:
- Mechanism of action: it acts by transferring **alkyl groups** to N7 position of **guanine** in DNA resulting in either **DNA strand breakage , cross linking of the two strands or intrastrand linking** so that normal synthesis is prevented. Dividing cells are more sensitive to the action of the drug. **Nitrogen mustard**

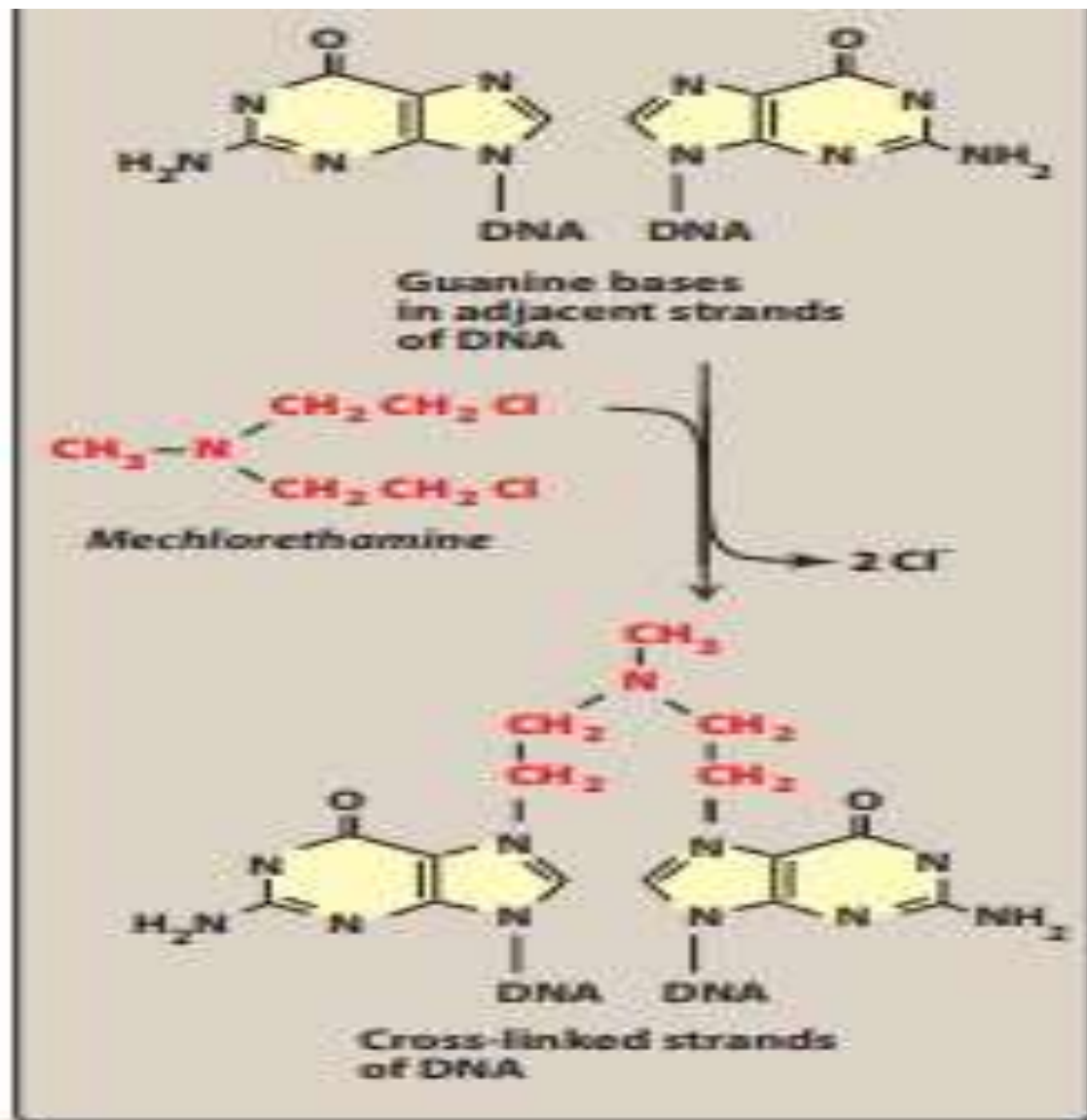
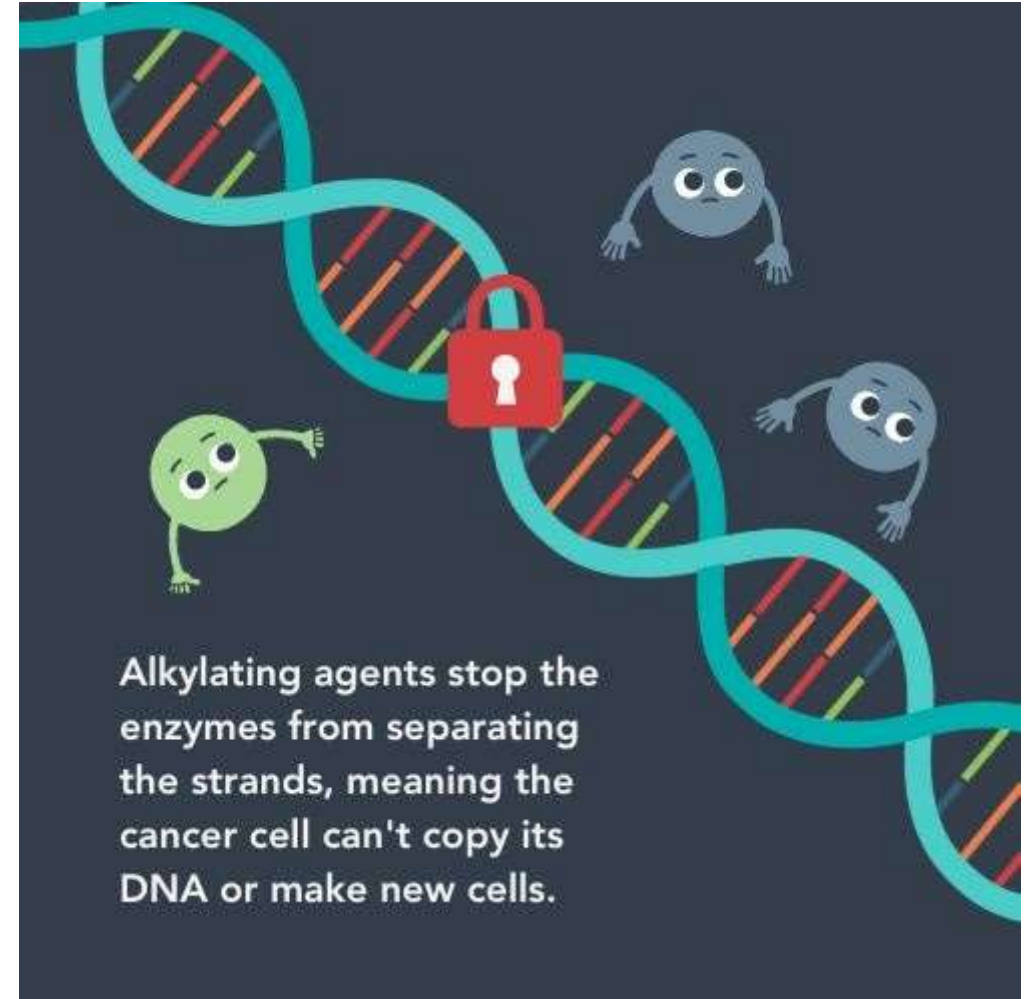


Figure 39.21

Alkylation of guanine bases in DNA is responsible for the cytotoxic effect of *mechlorethamine*.



Adverse effects:

- *Severe nausea and **vomiting** of central origin. This can be diminished by prior administration of **ondansetron**(serotonin 5-HT₃ receptor antagonists.) **and dexamethasone**.
- ***Bone marrow depression** is severe, latent viral infection (immunosuppressant) occurs.
- *If extravasation occurs, it must be treated with **sodium thiosulfite**.

B -Cyclophosphamide:

Orally administered or IV

Adverse effects:

- nausea, vomiting, diarrhea and myelodepression.
- Hemorrhagic cystitis is caused by **acrolein** *This toxicity can be diminished* by increasing fluid intake and by MESNA (sodium-2-mercaptoethane sulphonate) which interact with acrolein forming a non-toxic compound.
- **Amenorrhea, testicular atrophy and sterility.
- Veno-occlusive disease of liver, secondary malignancies.
- **Neurotoxicity

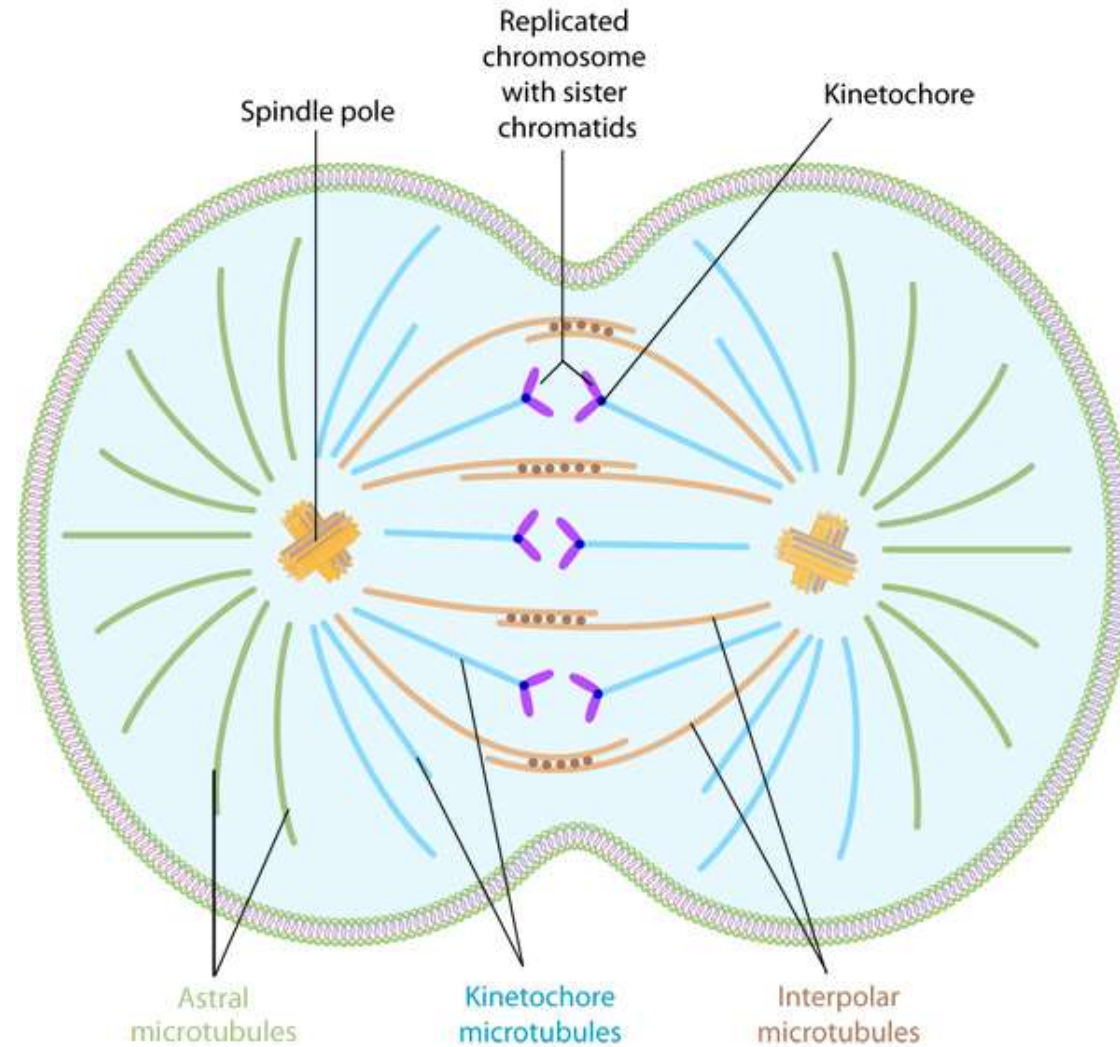
C -Nitrosureas:

- **Lomustine and carmustine** have the ability to pass BBB, thus used for treatment of **brain tumors**.
- **Streptozotocin** is used in **insulinomas** since it is specifically toxic to β cells of islets of Langerhans.
- **Mechanism of action:** They function by cross linking through **alkylation** of DNA inhibiting its replication→ inhibition of RNA and protein synthesis.
- **Adverse effects:** delayed myelodepression and aplastic anemia on prolonged use, renal and pulmonary toxicity.
- ***Streptozotocin is diabetogenic.***

IV-Microtubule inhibitors:

- Microtubules are important parts of cytoskeleton (needed for movements of cytoplasmic structures) and mitotic spindle (consisted of chromatides and microtubules composed of tubulin).
- The ***mitotic spindle is essential for equal portioning of DNA into two daughter cells during cell division.***
- These drugs act by affecting the equilibrium between polymerized and depolarized forms of microtubules.

Mitotic spindle



- A-Vincristine (VX) and vinblastin (VBL):

- Referred to as **vinca alkaloids**. VX is used in combination with other drugs for the treatment of **leukemias** (POMP), Wilms tumor, ... While VBL is used for treatment of testicular carcinoma with bleomycin and cisplatin.

- **Mechanism of action:**

- They are **CCS** and **phase** specific.

Adverse effects:

- GIT, alopecia, if extravasation occurs→ phlebitis or cellulites.
- VBL is more myelodepressant than VX, while VX causes **neuropathy (areflexia : absence of reflexes) and muscle weakness** especially the quadriceps.
- **Constipation** is the most common symptom of **autonomic neuropathy**..

- **B -Paclitaxel and docitaxel:**
- **Adverse effects:**
- ***Neutropenia*** which is prevented by granulocyte colony-stimulating factor (*G-CSF*)(*filgrastim*).
- Serious **hypersensitivity** reaction necessitates pretreatment with dexamethasone, diphenhydramine and H2 blocker.
- **Peripheral sensory neuropathy** and alopecia can occur.

V-Steroid hormones and their antagonists:

- Tumors that are steroid hormone sensitive may be either:

1-Hormone – responsive: in which tumor regresses *following giving* treatment with the specific hormone. This is only *palliative* except in case of lymphomas.

• 2-Hormone- dependent (sensitive) in which *removal* of hormonal stimulus *causes tumor regression*. Removal is achieved by surgery or drugs (tamoxifen in breast cancer).

A-Corticosteroids:

- **Prednisone:** it reduce the inflammation and help prevent an allergic reaction to cancer treatments. reduce your body's immune response, for example after a bone marrow transplant. help reduce sickness when having cancer treatment
- **Adverse effects:**
 - hyperglycemia, hypertension, iatrogenic Cushing syndrome, cataract, glaucoma, osteoporosis, depression, euphoria, psychosis, peptic ulceration and delayed healing of wounds (immunosuppressive),.....
- **Uses:** in lymphomas and leukemias.

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- B-sex hormones:
- **Estrogen and progestins** ; Estrogen can cause myocardial infarction , thromboembolic accidents , strokes, impotence ,... when used in treatment of cancer prostate.

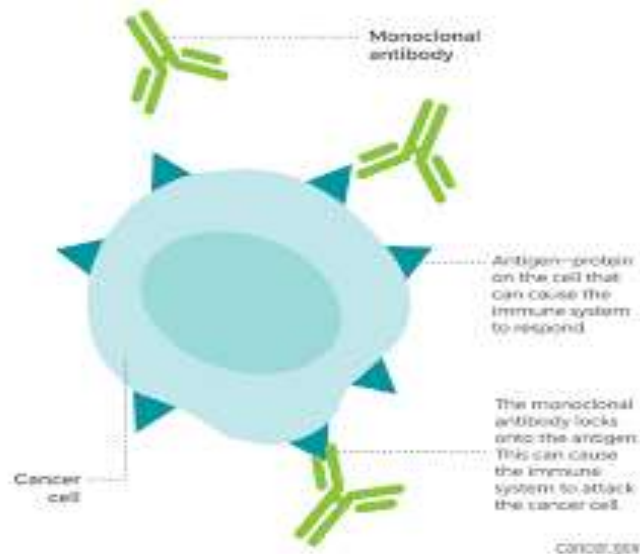
C- sex hormones antagonists:

- 1-Tamoxifen:- CCNS

- It is an anti estrogen drug with weak estrogenic activity.
- -A selective estrogen receptor modulator (SERM), also toremifene.
- -Used as 1st line treatment of estrogen receptor **positive breast cancer.**

VI-Monoclonal antibodies:

- *Directed at specific targets and produce fewer adverse effects.*
- **"Humanized"** ABs are now available by DNA recombinant technology by replacing part of mouse gene sequence with human genetic material.



VII-Other chemotherapeutic agents:

- a-Platinum coordination complexes:
- Cisplatin and Carboplatin
- Mechanism of action: similar to alkylating agents.
- They are **CC non specific**. They can bind proteins and compounds containing (-SH) groups.

Cont.,

- **Adverse effects:**

- severe **persistent vomiting** which is ameliorated by pretreatment with antiemetic drugs. **Nephrotoxicity** due to **Cisplatin's proximal tubular toxicity** which necessitates intensive hydration and diuresis. **Ototoxicity and neurotoxicity** also occurs and increased with concomitant administration of **aminoglycosides**
- **Alopecia due to outer hair cell toxicity.**

Care Capsule



B.Interferons (IFN):

- They are antineoplastic , antiviral and immunosuppressive .Classified into 3 types: α , β , γ .

- Adverse effects:

- **Depression** with alpha IFN. **Leukopenia** and thrombocytopenia are dose-limited toxicities.
- Fever and chills during 1st days of treatment.
- Elevation of **liver** enzymes may occur, also **nephrotoxicity** at high doses.
- **Loss** of weight, **hairs** and fatigue.
- Neuropathy and myopathy.